

Hector E. Delgado Diaz

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Education

- Sept 2019 – Aug 2026 **Ph.D. University of Washington**, Astronomy and Astrobiology
- M.S. in Astronomy degree conferred 2021-03
- Aug 2017 – May 2019 **M.S. California State University Los Angeles**, Physics
- Dissertation Title: Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water
- Aug 2012 – May 2017 **B.Sc. Universidad de Puerto Rico en Cayey**, General Sciences

Publications

The Effect of Stellar Energetic Particles from a Large M Dwarf Flare on the Atmospheric Chemistry of Potentially Habitable Exoplanets (in-prep)

H. Delgado Diaz, A. Miranda-Rosete, A.a Segura, G. Gronoff, V. Airapetian, J. Kasting
Monthly Notices of the Royal Astronomical Society

Climates of Gl 514 b (accepted)

H. Delgado Diaz, R. Barnes, R. Deitrick, M. Damasso
The Astrophysical Journal

- Feb 2026 **The Effect of Tidal Heating and Volatile Budgets on the Outgassed Atmosphere on 55 Cancri e**
B. Nguyen, L. Schaefer, F. Dai, **H. Delgado Diaz**
[10.3847/1538-4357/ae3f21](https://doi.org/10.3847/1538-4357/ae3f21) [🔗](#) (The Astrophysical Journal)
- Nov 2019 **Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water**
R. Hu, **H. Delgado Diaz**
[10.3847/1538-4357/ab4cea](https://doi.org/10.3847/1538-4357/ab4cea) [🔗](#) (The Astrophysical Journal)

Academic Experience

- Seattle, WA, USA **University of Washington**, Research Assistant
Sept 2019 – present
- VPlanet Group; Advisor: Rory Barnes
 - Using VPlanet, an open source code written in C, developed simulations of exoplanets orbiting M dwarf stars and studied their habitability based on stellar, orbital, rotational, and atmospheric parameters.
 - With Python, created simulations of exoplanetary atmospheric spectra and used Bayesian approach to simulate JWST observations of such atmosphere to assess habitability.
- Seattle, WA, USA **University of Washington**, Teaching Assistant
Sept 2019 – present
- Taught the Activity portion of Introduction to Astronomy for undergraduate students in-person and online.
 - Moderated discussions for 30+ students in the online course.
 - Migrated and helped develop the "Introduction to Astrobiology" course from in-person to the online Canvas platform during the COVID-19 pandemic.
- Seattle, WA, USA **University of Washington**, Instructor
Sept 2024 – Mar 2025
- Co-Instructor of Cosmology (ASTR 425) with Dr. Jessica Werk
 - Instructor of Pre-Major in Astronomy Program (ASTR 192)

Ciudad de Mexico, Mexico June 2024 – Sept 2024	Universidad Nacional Autonoma de Mexico , Research Assistant • Instituto de Ciencias Nucleares; Advisor: Antigona Segura • Used Python and Fortran to understand how energetic particles from solar flares could affect atmospheric nitrogen on exoplanets.
Palo Alto, CA, USA June 2023 – Sept 2023	Stanford University , Research Assistant • Earth and Planetary Sciences; Advisor: Laura Schaefer • Used MATLAB to add the physics of CO₂ outgassing from planetary interior on a magma ocean code.
Geneva, Switzerland June 2019 – Sept 2019	CERN , Research Intern • ATLAS Experiment; Advisor: Matt LeBlanc • Pivoted into experimental particle physics: used C++ to develop data processing pipelines for proton-proton collision events collected by the ATLAS detector at the Large Hadron Collider.
La Canada, CA, USA Aug 2018 – May 2019	NASA Jet Propulsion Laboratory , Visiting Research Student • Astrobiology Division; Advisor: Renyu Hu • Used MATLAB to calculate the effect of lightning on atmospheric nitrogen when bodies of water are present on the exoplanet around GJ 876.
Los Angeles, CA Jan 2018 – June 2018	California State University Los Angeles , Research Assistant • Physics Department; Advisor: Luca Ricci • Used ALMA radio telescope data to study the chemical components of protoplanetary disks.
Los Angeles, CA Aug 2017 – May 2019	California State University Los Angeles , Teaching Assistant • Assisted in PHYS 2100 and 2200 laboratory classes and taught the discussion portion of the course. • Held office hours and participated on the Physics and Astronomy tutoring center.
Cayey, PR, USA Aug 2015 – May 2017	Universidad de Puerto Rico en Cayey , Student Researcher • Laboratorio de Modelaje y Simulaciones Teoricas • Modeled rate of reaction for different configurations of atmospheric CO₂ and H₂O.
Greenbelt, MD, USA June 2016 – Aug 2016	NASA Goddard Space Flight Center , Summer Research Intern • Planetary Systems Laboratory; Advisor: Paul Romani • Used data from the Cassini-Huygens mission to constrain the source of stratospheric water of Saturn
Washington, DC, USA Jan 2015 – June 2015	The Washington Center , International Relations Intern • Social Inclusion Division at Organization of American States; Advisor: Alexandra Barrantes • Assisted in communications between the Division of Social Inclusion and Diplomats of the American States for diplomatic meetings. • Assisted in Translating, proofreading, and formatting the book "Protocolo San Salvador" • Live English scribe for the Secretary of Social Inclusion Meetings for the American States. • Completed course on International Markets and Business
San Jose, CA, USA June 2014 – Aug 2014	San Jose State University , Summer Research Intern • Laboratory of Neuroscience; Advisor: Katherine Wilkinson • Used mice leg muscles to study the effects of lactic acid on nociceptors III and IV neurons and how obesity could be related to associated muscular and joint pain.

Awards and Fellowships

Seattle, WA, USA Sept 2025	Wallerstein Graduate Student Research Fellowship
Santander, Spain July 2023	Beca Josep Comas i Sola Escuela de Astrobiologia
Apr 2021	Bonderman Fellowship Award • Selected out of 200 applicants to travel the world for 8 months as part of a cultural experience.
Seattle, WA, USA Sept 2020	Department of Astronomy Teacher of the Year Award
Seattle, WA, USA Aug 2019	ARCS Foundation Fellowship • Selected for a 3-year fellowship totaling \$17,500.
Seattle, WA, USA Aug 2019	GEM Fellow
Chur, Switzerland June 2019	Exo-Atmospheres Summer School Scholarship
Los Angeles, CA, USA Aug 2018	Hispanic Scholarship Fund Award

Outreach and Volunteering Experience

June 2026 – present	AAS Committe on Astronomy and Public Policy (CAPS) - Committee Member
Dec 2024 – present	Origins: Life in the Universe • Co-founder of the Astrobiology Registered Student Organization at the UW • Graduate Student Advisor position
Sept 2024 – present	UW Astronomy Graduate Senator • Member of the Diversity Committee, Travel Grants Committee, and U-Pass Advisory Board • Monthly meetings held for the General Senate and weekly meetings for each committee • Co-writer of approved resolutions to improve graduate student life at the UW
Feb 2022 – present	Raising eSTEAM • Co-founder of the initiative • Current Grant Coordinator position with the task of finding and applying to grants to fund the program. • Tutor and teach STEM courses to incarcerated youth to help mitigate the vicious cycle of crime and punishment
Oct 2021 – present	NASA Subject Matter Expert Volunteering
Oct 2019 – present	UW Planetarium • Volunteer presenter for public shows and school field trips at the UW Planetarium.
Feb 2025	Peer Reviewer, Astrophysical Journal Supplement Series • Invited to review a manuscript on distance estimates for RR Lyrae Stars
Aug 2023	VPLanet Workshop - Tech Support
Sept 2020 – May 2021	UW Astronomy Graduate Senator

Research Mentorship

- 2025 **Avery Wolf (University of Washington)**
- 2025 **Jessica Lemus (University of Washington)**
- 2025 **Nathaniel Brown (University of Washington)**
- 2024-2025 **Kyler Reynolds (Everett Community College)**

Work Experience

- Seattle, WA, USA **University of Washington Upward Bound**, High School Science and Astronomy Teacher
June 2025 – Aug 2025
- Designed and taught "Introduction to Astronomy and Astrobiology" to high school students.
 - Taught high school students how to communicate science through the use of infographics.
 - Designed a solar powered miniature car using circuits and TinkerCAD.
- Pasadena, CA, USA **Fusion Academy**, High School Science Teacher
June 2022 – Sept 2023
- Taught high school students Science courses: Astronomy, Chemistry, Physics, Biology, Math (Algebra-Calculus), and Earth and Space Science.
 - Served as a substitute teacher for Spanish and Economy.
- Los Angeles, CA, USA **Griffith Observatory**, Museum Guide and Online Support
June 2021 – Sept 2023
- Worked explaining astronomy history, instruments, and science to the general public in English and Spanish.
 - Worked as online support for the Griffith Museum Online School Program.

Talks and Posters

- Jan 2025 **Climates of Gl 514 b**
- American Astronomical Society 245th Winter Meeting, Seattle, WA, USA
- Aug 2024 **Climates of Gl 514 b**
- NASA Goddard: ROCKE-3D Journal Club
- Aug 2024 **Climates of Gl 514 b**
- Simposio en Materiales y Fisicoquímica
- Sociedad Dominicana de Física
- July 2023 **FLUXS - Gas Fluxes for Understanding and Experimenting the Subsurface**
- Mars Ideation Factory, NASA Goddard
- June 2023 **Climates of Gl 514 b**
- Qualifying Exam, University of Washington
- May 2023 **Climates of Gl 514 b**
- AbGradCon, San Diego, CA, USA
- Jan 2023 **Climates of Gl 514 b (poster)**
- American Astronomical Society 241st Winter Meeting, Seattle, WA, USA
- Dec 2021 **Procurando Exoplanetas Habitables con el Telescopio JWST**
- Spanish talk for the Museum of Flight, Seattle, WA, USA
- Oct 2021 **Searching for Habitable Exoplanets Using the Telescope JWST**
- Museum of Flight, Seattle, WA, USA
- May 2021 **Characterizing Exoplanetary Atmospheres with JWST**
- Town Hall Seattle, WA, USA

- Jan 2021 **Caracterizando Atmosferas de Exoplanetas con JWST**
 - Sociedad Dominicana de Fisica, presentacion en linea
- Jan 2019 **Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water**
 - American Astronomical Society 233rd Winter Meeting, Seattle, WA, USA
- Nov 2018 **Stability of Nitrogen in Planetary Atmospheres in Contact with Liquid Water**
 - American Physical Society Bridge Program Conference, Mountain View, CA, USA
- Oct 2016 **Saturn's Stratospheric Oxygen Compounds**
 - Division of Planetary Science, Pasadena, CA, USA
- Nov 2015 **Effect of Obesity and Lactic Acid on Muscle's Nociceptors III and IV neurons**
 - Annual Biomedical Research Conference for Minority Students, San Antonio, TX, USA
- Oct 2015 **Effect of Obesity and Lactic Acid on Muscle's Nociceptors III and IV**
 - Society for the Advancement of Chicanos and Native Americans in Science, Los Angeles, CA, USA

Professional Development

- Greenbelt, MD, USA **Mars Ideation Factory**
Aug 2023
- Santander, Spain **Astrobiology Summer School**
July 2023
- Pasadena, CA, USA **Carl Sagan Summer Workshop**
July 2023
- Chur, Switzerland **Exo-Atmospheres Summer School**
June 2019